

The 0.8 rung: a judgment

there is no gap to certify

PRIME.RDAGGER.L08_JUDGMENT.01 — round 502

Stefan Hamann

August 31, 2026

Abstract

Adjudication of the round-496 no-go at $L = 0.8$ and of its two named obstructions (cut margin $-1.61 \cdot 10^{-4}$; noncompact prime translations, off-block 0.9395, $3 \times \text{HS}$ rest 27.367). The r496 numbers are first reproduced from an independent implementation — all four cutoff rows, κ_w , $\|T_w\|_{\text{HS}} = 19.2177$, $r_{101} = 9.12241$ — which also identifies the booking r496 actually used (prime translations retained as operators). **U1 BLOCKED(no-certifiable-gap@ $L \geq 0.45$)**. Both obstructions are mis-specified, because the quantity they are trying to make positive does not exist at the claimed size. An explicit odd test function of Legendre degree ≤ 13 gives

$$\lambda_*(0.8) \leq 1.8170 \cdot 10^{-14},$$

verified in two independent representations: the x -space kernel form at $\text{dps}=30$ gives $1.81691694981 \cdot 10^{-14}$, and the spectral side of the explicit formula, summed over the first 60 zeros of ζ (all terms nonnegative), already gives $1.442289883 \cdot 10^{-14}$, i.e. 79% of it. The r496 obstruction margins are therefore ten orders of magnitude *above* the target. Measured ladder of ceilings: $7.57 \cdot 10^{-3}$ ($L = 0.30$), $1.33 \cdot 10^{-3}$ (0.34657), $1.81 \cdot 10^{-4}$ (0.40), $1.62 \cdot 10^{-5}$ (0.45), $9.34 \cdot 10^{-7}$ (0.50), $1.61 \cdot 10^{-9}$ (0.60), $4.87 \cdot 10^{-13}$ (0.70), below the float floor from $L = 0.75$; the archimedean floor meanwhile falls to -0.92648 at $L = 0.8$ and -12.518 at the r471 top rung $L_{16} = 2.7726$, so the three channels must cancel to 13.6 digits at $L = 0.8$ already. Four candidate refinements are calibrated against these numbers and all four die, each for a stated reason; the finest cutoff scan drives the r496 sign deficit to $-7.44 \cdot 10^{-11}$ and its limit is the ceiling itself. **U2 LADDER_RESTS / REDIRECT_TO_BRIDGE_LANE**: the λ_* ladder is closed as a proof route, permanently, at every rung beyond the prime-free zone. The proved r493e assembly **grid_dense_extension** consumes *non-strict* positivity uniform in the support length; a section-plus-remainder architecture can never produce a non-strict inequality. **U3** adds family F-VII (fixed- L margin objects read as proof targets) and six entries. **Free diagnosis**: the compressed prime translations are exactly $\frac{1}{2} \times$ the adjacency operator of a path graph on $\lceil 2L/\log n \rceil$ vertices, so $\text{spec}(\text{Re } R_{\log n}) = \{\cos(\pi j/(m_n + 1))\}$ — measured $\pm 1/\sqrt{2}$, $\pm \frac{1}{2}$, $\pm \frac{1}{2}$ at $L = 0.8$. Their obstruction is not noncompactness; it is incommensurability. No RH claim, no anti-RH claim.

Status. U1 BLOCKED(no-certifiable-gap@ $L \geq 0.45$) / U2 LADDER_RESTS_REDIRECT_TO_BRIDGE_LANE / U3 ANTI_LIST_EXTENDED. No promotion of an RH claim, no ledger row, no sealed pin.

Claim boundary. This is an adjudication note. It contains no probe, no verifier, no new census and no sealed pin; every numeral is a float or $\text{dps}=30$ re-measurement made to ground a verdict. The inequality $\lambda_*(0.8) \leq 1.8170 \cdot 10^{-14}$ is an *upper* bound on a quantity that RH predicts to be ≥ 0 ; a small positive λ_* is exactly what RH predicts, and an upper bound on it is **not** an anti-RH statement. Both representations return a *positive* value for the witness and they agree; nothing anomalous is observed. The r494/r495 theorem $\lambda_*(0.3) \geq 2.1 \cdot 10^{-3}$ and the r471/r477 finite-subspace certificates are untouched. No L^* claim, no $R^\dagger$ claim, no RH claim, no anti-RH claim.

1 Convention

$\text{supp } h \subset [-L, L]$, $g = h * \tilde{h}$, $g(0) = \|h\|_2^2$. With E the zero extension, $R_x = E^* \tau_x E$, $a_n = \Lambda(n)/\sqrt{n}$ and $k(x) = e^{x/2}/\sinh x$, the r494 form is

$$Q_W(h) = \underbrace{\int_0^{2L} k(x)(g(0) - g(x))dx - c_L g(0) + \Pi(h)}_{A(h) + \Pi(h)} - \underbrace{2 \sum_{n \leq e^{2L}} a_n g(\log n)}_{P(h)}, \quad (1)$$

$\Pi(h) = 2(C^2 - S^2)$, $C = \langle h, \cosh(\cdot/2) \rangle$, $S = \langle h, \sinh(\cdot/2) \rangle$, and $\lambda_*(L) = \inf\{Q_W(h)/\|h\|^2\}$ over the support class. Two bookings of the prime term recur below: (A) $-P \succeq -2 \sum a_n \cdot I$, dropping the positive translation differences; (B) $-P$ kept as the operator $-2 \sum a_n \text{Re } R_{\log n}$. The r469 scramble gate is live at this L (the nodes carry the literal $\log n$). The r492 anti-list, including pads N1–N4, is in force; in particular every Galerkin number below is typed as an **upper** bound on an operator eigenvalue, which is the only direction used.

2 What was re-measured

An independent implementation of (1) was written for this judgment (normalized Legendre Galerkin; the near-diagonal integrand assembled through the cancellation-free split $2(g(0) - g(x)) = \int_{-L}^{L-x} |h(y) - h(y+x)|^2 dy + \int_{-L}^{-L+x} |h|^2 + \int_{L-x}^L |h|^2$).

2.1 Calibration against sealed numbers

Quantity	this note	sealed	round
$c_L(0.3)$	2.1924049111319992553	2.1924049111319992553	r495
$\kappa_w(0.3)$, $[0.01, 0.03]$	3.6980080402989774	3.6980080402989775	r494
odd section, $L = 0.3$	$2.2256359052 \cdot 10^{-1}$	$2.2256359052 \cdot 10^{-1}$	r494
first Dirichlet, $L = 0.3$	$1.18025311091851 \cdot 10^{-2}$	$1.18025311091317 \cdot 10^{-2}$	r494
$c_L(0.8)$	3.5598735700355447	3.5598735700355446	r496
$2 \sum a_n$, $L = 0.8$	2.9419735252236205	2.9419735252236205	r496
constant mode, $L = 0.8$	0.0802740549593	$[0.0802740548742, 0.0802740551870]$	r477

2.2 Reproduction of the r496 no-go

$[x_0, x_1]$	κ_w	$\lambda_{\min}(P_{101} Q_W^{(w)} P_{101})$	r496	$\ T_w\ _{\text{HS}}$
$[0.001, 0.003]$	7.3770530633	$-1.610673 \cdot 10^{-4}$	$-1.6106733 \cdot 10^{-4}$	19.2177
$[0.002, 0.006]$	6.6829061429	$-8.176622 \cdot 10^{-4}$	$-8.1766226 \cdot 10^{-4}$	13.5946
$[0.003, 0.010]$	6.1992956339	$-2.284102 \cdot 10^{-3}$	$-2.2841017 \cdot 10^{-3}$	10.6441
$[0.010, 0.030]$	5.0654766992	$-3.032258 \cdot 10^{-2}$	$-3.0322577 \cdot 10^{-2}$	6.0742

All four rows, all four κ_w , and $r_{101} = 9.12241$ against the sealed 9.122410180 (hence $3r_{101} = 27.3672$ against 27.367230540) reproduce. **Booking identified:** the reproduction only matches under booking (B). Under booking (A) the same sections give -3.8688 , four orders of magnitude lower. This matters: r496’s surviving margin is the one that keeps the prime translations *in* the error block, which is exactly why its G3 then failed.

3 U1 — the ceiling

3.1 The witness

The uncut Legendre Galerkin $\lambda_{\min}$ of (1) at $L = 0.8$ is $6.07 \cdot 10^{-15}$ at $n = 101$ and does not move with n ($7.79 \cdot 10^{-14}$, $8.73 \cdot 10^{-15}$, $1.10 \cdot 10^{-14}$, $7.26 \cdot 10^{-15}$ at $n = 25, 51, 101, 151$): the value is at

the float floor, not converging to anything resolvable. Its eigenvector is odd to $1.5 \cdot 10^{-8}$ of its mass, with dominant normalized Legendre coefficients

$$(c_1, c_3, c_5, c_7, c_9, c_{11}, c_{13}) \approx (0.4099, 0.6641, 0.5495, 0.2847, 0.0891, 0.0103, 0.00456),$$

and 99.9999% of its mass below degree 20. Evaluated as an explicit function — one polynomial, no Galerkin, no eigenvalue — in the x -space form at **dps=30**:

$$\frac{Q_W(h)}{\|h\|_2^2} = 1.81691694981 \cdot 10^{-14}$$

against channel values $A(h) = -0.24300$, $\Pi(h) = -0.02894$, $-P(h) = +0.27194$: a 13.6-digit cancellation.

3.2 The independent confirmation

Under (1) the program's own normalization makes $Q_W(h) = \sum_\rho \hat{g}(\rho)$, which on the critical line is $\sum_\gamma |\hat{h}(\gamma)|^2$ — a sum of *nonnegative* terms. Every partial sum is therefore a hard lower bound, and it is a third representation, independent of both the x -space and the frequency form. With the first 60 zeros of ζ ($\gamma_{60} = 163.0307$):

Test function	x -space $Q_W/\ h\ ^2$	spectral, 60 zeros	captured
$L = 0.8$ constant (r477 control)	0.0802740549593	0.07050676638	88%
$L = 0.3$ even minimiser (r494 control)	0.0075718518899	0.007040224407	93%
$L = 0.6$ witness	$1.62318271558 \cdot 10^{-9}$	$1.39688596 \cdot 10^{-9}$	86%
$L = 0.8$ witness	$1.81691694981 \cdot 10^{-14}$	$1.442289883 \cdot 10^{-14}$	79%

The two controls behave as they must (monotone from below, under the total). The $L = 0.8$ witness is bracketed: $1.4423 \cdot 10^{-14} \leq Q_W(h)/\|h\|^2 = 1.8169 \cdot 10^{-14}$. Term by term the low zeros are minuscule — the γ_1 term is $1.06 \cdot 10^{-26}$, i.e. $|\hat{h}(\gamma_1)| \approx 7.3 \cdot 10^{-14}$ against $\|\hat{h}\|_2^2 = 2\pi$. The near-null direction is real, not a numerical artefact, and it is a genuine near-annihilator of the low zeros.

Proposition 1 (the ceiling). $\lambda_*(0.8) \leq 1.8170 \cdot 10^{-14}$, *witnessed by one explicit odd polynomial. Sealing it needs one interval quadrature of that polynomial; it is not sealed here and the verdict does not need it.*

3.3 The ladder

Galerkin ceilings ($n = 81$; upper bounds on λ_* , which is the correct logical type) against the archimedean-only floor $\lambda_{\min}(A + \Pi)$, and the resulting cancellation depth at the witness:

L	nodes	$\lambda_{\min}(A + \Pi)$	$\lambda_*(L) \leq$	digits
0.30000	0	$+7.5715 \cdot 10^{-3}$	$7.5715 \cdot 10^{-3}$	2.1
0.34657	0	$+1.3294 \cdot 10^{-3}$	$1.3294 \cdot 10^{-3}$	2.9
0.40000	1	$-7.8049 \cdot 10^{-2}$	$1.8136 \cdot 10^{-4}$	3.8
0.45000	1	$-2.0521 \cdot 10^{-1}$	$1.6166 \cdot 10^{-5}$	4.9
0.50000	1	$-3.2224 \cdot 10^{-1}$	$9.3448 \cdot 10^{-7}$	6.1
0.60000	2	$-5.3588 \cdot 10^{-1}$	$1.6096 \cdot 10^{-9}$	8.9
0.70000	3	$-7.3394 \cdot 10^{-1}$	$4.8745 \cdot 10^{-13}$	12.5
0.80000	3	$-9.2648 \cdot 10^{-1}$	float floor	13.6
1.00000	5	-1.3241	float floor	—
1.50000	12	-2.6504	float floor	—
2.77260	70	-12.5178	float floor	—

Three readings, all load-bearing for the verdict.

1. **The collapse is ≈ 3 decades per $\Delta L = 0.1$** and it starts exactly where the prime nodes turn on. By $L = 0.45$ the required relative precision is already below 10^{-5} ; by $L = 0.7$ it is $10^{-12.5}$.
2. **The archimedean channel goes negative at $L \approx 0.37$** and reaches -0.92648 at $L = 0.8$. Beyond the prime-free zone the prime term is not a perturbation to be bounded — it *carries* the positivity, and it must do so to the last digit.
3. **r494 succeeded because $L = 0.3$ needs 2.1 digits.** That is the whole difference between r494 and r496, and it was invisible in the r492 ranking, which listed “same machine up the r471 schedule, 2–5 rounds”. That ranking entry is hereby withdrawn.

3.4 The candidate table

Each candidate was calibrated against the numbers above *before* being named or killed, per the r492 standard.

Cand.	Verdict	Ground
(K1) finer cutoff window	KILL, transposed	the deficit really does vanish — and its limit is t (§3.5)
(K2) prime-free error block via a scalar	KILL $\times 2$	$\lambda_{\min}(A + \Pi) = -0.92648 < 0$ kills every scalar or booking (§3.6)
(K3) primes as an exact finite-parametric object	KILL, structural	right per node, impossible jointly: $\log 3 / \log 2 \notin$
(K4) strip-booked Loewner cutoff	KILL at the witness	the witness has $h(\pm L)^2 = 9.84 \cdot 10^{-14}$; the st nothing (§3.8)

3.5 (K1) The cutoff scan

The r496 window grid had four rows. Continuing it:

$[x_0, x_1]$	κ_w	$\lambda_{\min}(P_{101}Q_W^{(w)}P_{101})$	r_{101}
$[10^{-3}, 3 \cdot 10^{-3}]$	7.3771	$-1.6107 \cdot 10^{-4}$	9.12
$[3 \cdot 10^{-4}, 10^{-3}]$	8.5048	$-1.3454 \cdot 10^{-5}$	21.02
$[10^{-4}, 3 \cdot 10^{-4}]$	9.6805	$-9.0685 \cdot 10^{-7}$	48.16
$[3 \cdot 10^{-5}, 10^{-4}]$	10.8077	$-9.2299 \cdot 10^{-8}$	94.96
$[10^{-5}, 3 \cdot 10^{-5}]$	11.9832	$-6.9242 \cdot 10^{-9}$	182.65
$[10^{-6}, 3 \cdot 10^{-6}]$	14.2858	$-7.4437 \cdot 10^{-11}$	601.58

So O1 is not a wall: the sign deficit falls monotonically and would cross zero. It crosses at the uncut value — the ceiling of Proposition 1. The scan is converging to the right answer and the right answer is 10^{-14} . Quantitatively, at the witness the booked cutoff loss is dominated by its Dirichlet part, $\frac{1}{4}x_0^2\|h'\|^2$ with $\|h'\|^2 = 34.5442$, i.e. $8.636x_0^2$. For a certificate to remain positive *at the witness* it must satisfy $8.636x_0^2 < 1.8170 \cdot 10^{-14}$, hence

$$x_0 < 4.59 \cdot 10^{-8} \implies n \gtrsim 2L/x_0 = 3.5 \cdot 10^7, \quad \|T_w\|_{\text{HS}} \approx 2.8 \cdot 10^3,$$

and $3r_n < 1.8 \cdot 10^{-14}$ then demands a relative Hilbert–Schmidt tail of $2 \cdot 10^{-18}$, pushing n higher again. Against r494’s $401^2 = 1.6 \cdot 10^5$ interval-certified entries this is $\gtrsim 10^{15}$ of them. **Kill, transposed:** the mechanism is sound, the target is not there.

3.6 (K2) A prime-free error block

This is O2's own question — can (1) be split into an archimedean part certifiable the 0.3 way, plus prime Loewner positivity, minus a prime scalar, with the error block prime-free? Two sharpenings of the scalar were found and measured, and both are irrelevant for the same reason.

The almost-periodic sup. The crude scalar $2 \sum a_n = 2.9420$ is not the sharp one. Since compression never increases a multiplier's sup,

$$-P \succeq -\pi_L I, \quad \pi_L = \sup_t \left(-2 \sum_{n \leq e^{2L}} a_n \cos(t \log n) \right).$$

At $L = 0.8$ this is available in closed form, because $\log 4 = 2 \log 2$: with $c = \cos(t \log 2)$ the $n \in \{2, 4\}$ part is the quadratic $a_2 c + a_4(2c^2 - 1)$, minimal at $c = -a_2/(4a_4) = -1/(2\sqrt{2})$, while $\cos(t \log 3) = -1$ is attainable in the closure because $\log 3 / \log 2 \notin \mathbb{Q}$ (Weyl). Hence $\pi_{0.8} = 2.13500217689506$, confirmed by a brute-force scan (2.134999 over $t \leq 4000$): a gain of 0.8070 over $2 \sum a_n$.

The boundary-strip minorant. Better still, from the exact zero-extension identity, $\|h - \tau_a h\|^2 \geq \int_{-L}^{-L+a} |h|^2 + \int_{L-a}^L |h|^2$, so

$$-P \succeq -\left(\sum_n a_n\right)I + M_V, \quad \sum_n a_n = 1.4709867626,$$

with M_V multiplication by an explicit nonnegative step function, levels (0.4907286, 0.9808577, 0.3465736, 0) from the centre outwards. This beats π_L and is exact.

Why both are dead. Any such split bounds the certificate by $\lambda_{\min}(A + \Pi) - (\text{prime scalar}) + \min V$. Measured, $\lambda_{\min}(A + \Pi) = -0.9264812$ at $L = 0.8$ (converged: $-0.9264551, -0.9264781, -0.9264812, -0.9264823$ at $n = 25, 51, 101, 151$), $\min V = 0$ near the boundary, and $\sup V = 0.9809 < \sum a_n$. Every scalar or potential booking of the prime term therefore returns a certificate between -2.40 and -3.87 ; the crude end of that range is exactly the measured booking-(A) section -3.8688 . There is a second, independent kill: even a perfect such split would have to land below $1.82 \cdot 10^{-14}$. **Kill** $\times 2$. The answer to O2's decomposition question is a measured *no*: beyond the prime-free zone the error block cannot be made prime-free, because the primes are not an error.

3.7 (K3) The primes as a finite-parametric object

O2's second intuition — three nodes, rank-like structure in the right coordinates — is correct per node, and is worth recording exactly.

Lemma 1 (compressed translations are path graphs). *Let $a > 0$, $m = \lceil 2L/a \rceil$. Then R_a is a partial isometry with $R_a^j = R_{ja}$ and $R_a^m = 0$, and $\text{Re } R_a$ is unitarily $\frac{1}{2}$ times the adjacency operator of the path graph on m vertices, tensored with the multiplicity of the cell shift. Hence $\text{spec}(\text{Re } R_a) \subset \{\cos(\pi j / (m + 1))\}_{j=1}^m \cup \{0\}$.*

Measured at $L = 0.8$, $n = 101$: $\text{Re } R_{\log 2}$ ($m = 3$) has $\lambda_{\max} = 0.707107 = \cos(\pi/4)$, $\text{Re } R_{\log 3}$ and $\text{Re } R_{\log 4}$ ($m = 2$) have $\lambda_{\max} = 0.500000 = \cos(\pi/3)$, each to all printed digits. So each prime block is bounded, noncompact, and *exactly diagonalizable in closed form*: r496's off-block 0.9395 is not a smallness failure, and calling these terms “noncompact” understates what is true of them individually.

The obstruction is joint. An exact finite reduction handling all nodes at once needs one cell length dividing every $\log n$; for $n = 2, 3$ that requires $\log 3 / \log 2 \in \mathbb{Q}$. It is not. What is left is the finite-section theory of an almost-periodic symbol on an interval — precisely the object whose frequency-side incarnation r492 closed permanently. **Kill, structural**, and this is the correct diagnosis of the r496 measurement.

3.8 (K4) The strip-booked cutoff

The refined Loewner booking O1 asks for exists:

$$\int_0^{x_0} k(x)(I - \operatorname{Re} R_x)dx \succeq \frac{1}{2}M_W, \quad W(y) = \int_{d(y)}^{x_0} k, \quad d(y) = \operatorname{dist}(y, \{\pm L\}),$$

worth $\approx x_0 h(\pm L)^2$, which for a generic mode at $x_0 = 10^{-3}$ is $\approx 6 \cdot 10^{-4}$ and would indeed dominate the $-1.61 \cdot 10^{-4}$ deficit. Two independent reasons it is worthless here. First, mechanically: M_W is an unbounded positive multiplication operator, and the block bound cannot digest it ($\|PM_W Q\|$ is not small); the best finite-rank minorant supported on the two inner half-strips carries only $\int_{x_0/2}^{x_0} k = \log 2$ and lands at $2.2 \cdot 10^{-4}$, marginal even against the wrong target. Second, and decisively: *the witness has no endpoint mass*. Measured, $h(L) = 3.14 \cdot 10^{-7}$, i.e. $h(\pm L)^2 = 9.84 \cdot 10^{-14}$ — the same order as the ceiling itself. The near-null direction essentially vanishes at the endpoints, so the entire strip booking is worth 10^{-13} there. **Kill at the witness.**

3.9 The verdict

Proposition 2 (round-502 judgment). *Beyond the prime-free zone the r492/r494 architecture — booked Loewner cutoff, Legendre finite section, remainder charged at $r + 2r$ — is **BLOCKED**. It certifies statements of the form $\lambda_*(L) \geq c > 0$ by subtracting three strictly positive booked quantities from a scalar balance. At $L = 0.8$ the target is $\leq 1.8170 \cdot 10^{-14}$, the finest booking measured anywhere in the (x_0, n) plane is $7.44 \cdot 10^{-11}$, and forcing the cutoff loss alone under the target requires $n \gtrsim 3.5 \cdot 10^7$ modes. From $L \approx 0.45$ upwards the required relative precision exceeds five digits and grows by ≈ 3 decades per $\Delta L = 0.1$; from $L \approx 0.75$ the target is below double precision. No refinement of the booking changes this, because the deficit is not in the booking.*

4 U2 — the resource decree

4.1 The ladder was asking the wrong question

The decisive observation is not numerical. The r493e Lean assembly

$$\text{grid_dense_extension} : (\forall f. 0 \leq \text{weilForm } f) \Rightarrow \forall F \text{ admissible. } 0 \leq \text{fullWeilForm } F$$

is proved sorry-free, and its hypothesis is **non-strict** positivity on the grid class, uniform in the support length. That is what the assembled proof consumes. A fixed- L gap $\lambda_*(L) \geq c > 0$ is therefore both *stronger* than what the assembly needs at that L and *weaker* than what RH needs, since RH needs all L . And a section-plus-remainder architecture can never deliver the non-strict statement: every remainder charge it makes is strictly positive, so it can only ever prove a strict inequality — exactly the kind that §3 shows does not survive. The λ_* ladder is thus not merely hard beyond $L = 0.3465$; it is aimed off-target.

4.2 Decree

1. **The λ_* ladder is closed as a proof route**, permanently, for every $L > (\log 2)/2$. No round may reopen a booked cutoff, a Legendre finite section with an HS remainder charge, or any variant of the r494 three-gate contract at a rung beyond the prime-free zone. Its legacy is the ceiling curve of §3, which is a genuine and reusable measurement.
2. **r494/r495 stand.** $\lambda_*(0.3) \geq 2.1 \cdot 10^{-3}$ is untouched, doubly checked, and now also *explained*: $L = 0.3$ is the last rung needing only 2.1 digits.
3. **Resources go to the bridge lane** (r497–r501: [2c] spectral-sum analysis, then [2d] the contour form). It is the long pole, it is producing at a steady cadence, and it is the lane whose output the assembly actually consumes. Nothing in this note touches it.

4. **Any future internal analysis round** must target the non-strict, L -uniform grid-class statement $0 \leq \text{wellForm } f$ by an exact or structural route — a sum of squares, a factorization, an identity — never a numeric margin.
5. **Feasibility pre-gate, mandatory.** Before any scheme is proposed at a support length L , the cancellation depth at L must be measured first (the “digits” column of §3). A scheme whose booked losses exceed $10^{-\text{digits}}$ is refused without further review. This gate would have refused r496 before it ran.
6. **F2b** (r471 density plus continuity modulus) is superseded for the fixed-support direction by the proved r493e density bridge, and is not revived by this note.
7. The r469 scramble gate stays in force; the measurements here are node-literal and scramble-sensitive.

5 U3 — anti-list

Families F-I-F-VI and pads N1–N4 of r492 stay in force. N1 is not violated here: Galerkin numbers are used only as upper bounds on λ_* , which is their correct logical type, and the load-bearing ceiling is a single high-precision Rayleigh quotient bracketed from below by a nonnegative spectral partial sum. New family:

Family	Rounds	Members
F-VII fixed- L margin objects read as proof targets	r471, r477, r492, r496, r502	see N5–N10
N5	a fixed- L finite-subspace value read as an approximation to $\lambda_*(L)$. Measured: the r471 $L = 0.8$ hint $2.645 \cdot 10^{-4}$ and $\mu_{10} = 1.3225 \cdot 10^{-4}$ exceed $\lambda_*(0.8)$ by more than ten orders of magnitude. They remain valid as subclass statements; r496 said so, and this is the number	
N6	any cutoff, window or booking margin quoted at $L > 0.4$ without the measured λ_* ceiling beside it. Type specimen: the r496 reading of $-1.61 \cdot 10^{-4}$ as “agonizingly close” — the sign deficit is real and it is 10^{10} times the target	
N7	the r492 U2-ranking entry “same machine up the r471 schedule, 2–5 rounds, yield $\lambda_*(L) \geq c$ at $L = 0.8$ and beyond” — refuted and withdrawn	
N8	the prime term booked as a scalar ($2 \sum a_n$), as an almost-periodic sup (π_L), or as scalar-plus-potential ($\sum a_n$ with M_V). All three give negative certificates, because $\lambda_{\min}(A + \Pi)(0.8) = -0.92648 < 0$	
N9	the compressed prime translations treated as a joint finite-rank or finite-section object. Per node they are exactly $\frac{1}{2} \text{adj}(P_{m_n})$ (Lemma 1); jointly they are incommensurable, $\log 3 / \log 2 \notin \mathbb{Q}$	
N10	a strict-gap architecture applied to a non-strict target. Every remainder charge is positive, so such an architecture cannot prove $Q_W \geq 0$ without a gap — and beyond $L \approx 0.75$ no gap survives double precision	

6 Free diagnoses

1. **Lemma 1** (exact spectrum of the compressed translation) is new here and is reusable wherever a truncated shift appears; it replaces “bounded but noncompact” with a closed form.
2. $\lambda_*(L)$ is the frame lower bound of the ζ -zero set for PW_L . Under (1), $\lambda_*(L) = \inf \sum_\gamma |\widehat{h}(\gamma)|^2$ over unit-norm type- L functions. The ceiling curve of §3 is the program’s first quantitative picture of that frame constant, and it explains the collapse: once the type is large enough, the low zeros can be near-annihilated ($|\widehat{h}(\gamma_1)| \approx 7 \cdot 10^{-14}$ at $L = 0.8$). Nothing about this is evidence for or against RH; it is the reason a gap-based route cannot climb.

3. **Cross-representation health.** The x -space form, the frequency form (r496 G1, $5.5 \cdot 10^{-14}$) and now the spectral form agree on every test function tried, including two controls pinned in earlier rounds. The witness value is positive in both representations that can produce a number. No anomaly.
4. **The r496 booking.** Reproduction shows r496's surviving row is booking (B). Its G2/G3 tension was therefore structural from the start: (B) is the only booking with a chance of positivity, and (B) is the only booking that puts noncompact translations in the error block. The two obstructions O1 and O2 were the same obstruction seen twice.

7 What is not claimed

No $\lambda_*(L) \geq c$ for any $L > 0.3$. No claim that $\lambda_*(0.8) = 0$: the measurement is an upper bound $1.8170 \cdot 10^{-14}$ and both representations return a positive value. No counterexample, no anti-positivity statement, and in particular nothing against RH — a small positive frame constant is what RH predicts. No Galerkin value is promoted to an operator value. The r471/r477 subclass certificates and the r494/r495 fixed-support theorem stand unchanged. Every numeral here is a float or **dps=30** re-measurement, not a sealed pin. No L^* claim, no $R^\dagger$ claim, no RH claim, no anti-RH claim.